Spatiotemporal Traffic Speed Forecasting: Applied Machine Learning Baselines, Spatial Graph Attention, and Chronological Benchmarking

Arun Kumar Giri*, Gaurav Meena†, Mausam Kumar Dwivedi‡, Ritik Atri§, Daniel Tayal¶
Department of Computer Science and Artificial Intelligence (CSAI)
Newton School of Technology, Rishihood University, Sonipat, India
Project Group: Team Quanta
Emails: *arun.giri2024@nst.rishihood.edu.in, †gaurav.meena2024@nst.rishihood.edu.in,
‡mausam.dwivedi2024@nst.rishihood.edu.in, §ritik.attri2024@nst.rishihood.edu.in, ¶daniel.tayal2024@nst.rishihood.edu.in

***Abstract*—Short-term traffic speed forecasting is a cornerstone of modern Intelligent Transportation Systems (ITS), dynamic congestion routing, and adaptive signal policy simulation. In this work, we investigate whether spatial graph representation learning enhances network-wide speed forecasts over independent road segment models. Using high-resolution five-minute sensor observations from the METR-LA (207 sensors, 34,272 timestamps) and PEMS-BAY (325 sensors, 52,116 timestamps) benchmarks, we implement a strictly causal, chronological evaluation protocol (70% train, 10% validation, 20% test). We formulate and compare four distinct paradigms across 5, 15, 30, and 60-minute horizons: a naive persistence baseline, segment-wise multi-output Random Forests, pooled sensor-aware XGBoost, and a spatial Graph Attention Network (GAT) operating over a Gaussian distance-thresholded sensor adjacency matrix. Missing observations and sensor outages are explicitly masked rather than imputed with zero speeds. Evaluated on 1,244,846 valid test target points, our empirical findings reveal that segment-wise Random Forests achieve superior accuracy across all horizons (MAE of 2.519 at 5 min to 4.192 at 60 min), outperforming pooled XGBoost and standalone spatial GAT. We present full performance breakdowns across all, peak, and off-peak periods, provide rigorous mathematical formulations of features and loss metrics, and analyze the architectural trade-offs between localized tabular models and graph neural representations.**

***Index Terms*—Traffic speed forecasting, intelligent transportation systems, random forests, XGBoost, graph attention networks, chronological backtesting, sensor outages.**

I. Introduction

Real-time traffic congestion imposes severe societal, economic, and environmental costs on metropolitan road networks. Accurate short-term traffic forecasting empowers municipal control centers to anticipate bottlenecks, optimize dispatch corridors, and coordinate signal timing [1]. However, urban traffic dynamics exhibit complex spatial autocorrelations governed by physical road network topology, coupled with non-stationary temporal periodicities reflecting diurnal commuting patterns and unexpected localized incidents.

Historically, traffic engineering relied on univariate statistical baselines such as Historical Averages (HA) and Autoregressive Integrated Moving Average (ARIMA) models [2]. While computationally lightweight, these formulations assume linearity and struggle with nonlinear spatiotemporal shockwaves. The advent of Applied Machine Learning (AML) introduced powerful non-parametric models such as Support Vector Regression (SVR) [3], Random Forests [4], and Extreme Gradient Boosting (XGBoost) [5]. These tree ensembles capture complex nonlinear lag relationships and calendar interactions effectively. Nonetheless, independent segment models treat spatial locations in isolation, failing to model how congestion propagates through upstream and downstream links.

Recently, Deep Learning architectures combining Graph Neural Networks (GNNs) with attention mechanisms or temporal convolutions have emerged to model network-wide road graphs [6]–[8]. Graph Attention Networks (GAT) dynamically assign learned attention coefficients to adjacent road links, theoretically enabling dynamic spatial information exchange.

A. Problem Statement and Objectives

Under the CSAI Project 4 mandate (*Traffic Flow Control via Graph Attention Networks and Flow Models*), this investigation addresses the following core questions:

1) Does spatial graph learning via GAT systematically outperform feature-engineered AML baselines (Random Forest, XGBoost) under identical, fair chronological evaluation?
2) How does predictive performance degrade across multiple forecast horizons ($h \in \{5, 15, 30, 60\}$ minutes) and under distinct traffic regimes (weekday peak vs. off-peak)?
3) How should sensor outages and zero-speed values be treated mathematically to prevent severe evaluation bias?

B. Key Contributions

The main contributions of this study are:

- **Strict Chronological Protocol**: We establish a leakage-free 70/10/20 chronological partition, ensuring standardizers are fit solely on training timestamps and forward sequence contamination is eliminated.
- **Causal Outage Masking**: We prove that zero-speed sensor readings represent sensor communication failure rather than genuine zero velocities; our loss functions

strictly mask missing targets while maintaining causal input integrity.

- **Comprehensive Multi-Horizon Benchmark**: We evaluate Persistence, Segment-wise Random Forest, Pooled XGBoost, and GAT across 1,244,846 valid test points, reporting MAE, RMSE, and MAPE across all, peak, and off-peak periods.
- **Network-Aware AML Formulations**: We engineer 1-hop spatial neighbor aggregations and rolling window statistics, bridging tabular feature engineering with graph topology.

II. Related Work

A. Statistical and Classical Baselines

Early traffic prediction frameworks primarily modeled univariate temporal series. Williams and Hoel [2] established theoretical foundations for seasonal ARIMA (SARIMA) processes in vehicular traffic, demonstrating strong performance for steady-state flows. Vector Autoregression (VAR) extended this to low-dimensional spatial clusters but suffered from the curse of dimensionality when scaled across hundreds of freeway sensors.

B. Applied Machine Learning Tree Ensembles

To address non-linearities, machine learning regressors became the benchmark for intelligent transportation systems. Castro-Neto et al. [3] demonstrated the robustness of Support Vector Regression under non-recurrent congestion. Tree ensembles, specifically Random Forests [4] and Gradient Boosted Decision Trees (XGBoost) [5], demonstrated superior performance on tabular representations by constructing orthogonal decision boundaries over lagged speed inputs, rolling moments, and cyclical calendar encodings.

C. Spatiotemporal Graph Neural Networks

Urban road networks possess irregular non-Euclidean graph topologies where standard Convolutional Neural Networks (CNNs) are unsuitable. Li et al. [6] proposed Diffusion Convolutional Recurrent Neural Networks (DCRNN), modeling traffic flow as a directed diffusion process. Yu et al. [8] developed Spatio-Temporal Graph Convolutional Networks (STGCN), fusing spectral graph convolutions with gated temporal units. Veličković et al. [7] introduced Graph Attention Networks (GAT), enabling nodes to assign anisotropic attention weights to their neighbors. In this work, we deploy spatial GAT over a fixed road-distance graph and benchmark its predictive fidelity against AML tree models.

III. Dataset Quality, Topology, and Protocol

A. Traffic Observation Benchmarks

Experiments are conducted on the official METR-LA and PEMS-BAY datasets commonly utilized in spatial forecasting literature:

- **METR-LA**: Recorded by 207 loop detectors on Los Angeles County freeways over 4 months (March 1, 2012 to June 27, 2012), comprising 34,272 continuous 5-minute time steps.
- **PEMS-BAY**: Recorded by 325 sensors in the California Bay Area over 6 months (January 1, 2017 to June 30, 2017), yielding 52,116 time steps with geographic metadata.

Importantly, both benchmarks record average vehicular *speed* (miles per hour). Traffic volume (flow) and lane occupancy are unavailable in the source data and are not synthetically fabricated.

B. Sensor Outage Identification and Masking

Sensor hardware frequently suffers transmission loss, hardware resets, or power dropouts. In METR-LA, 8.11% of all readings are recorded as exactly zero or NaN. Treating zero speeds as true physical vehicle halts distorts mean calculations and misleads gradient updates.

We formally define a binary observation mask $m_{s,t}$ for sensor $s \in \{1, \dots, S\}$ at time t:

$$m_{s,t} = \begin{cases} 1, & \text{if } v_{s,t} > 0 \text{ and } v_{s,t} \in \mathbb{R} \\ 0, & \text{otherwise (outage/missing)} \end{cases} \tag{1}$$

For causal model inputs, missing values are forward-filled using the most recent valid observation, with training-period median speeds serving as initial cold-start fallbacks. For training loss calculation and evaluation metrics, all entries with $m_{s,t} = 0$ are strictly masked.

C. Road Network Graph Construction

The spatial sensor topology is constructed from pairwise driving distances obtained from the official road network geometry. Let $d(i, j)$ denote the road distance between sensor i and sensor j. Following the thresholded Gaussian kernel formulation [6]:

$$W_{ij} = \exp\left(-\frac{d(i,j)^2}{\sigma^2}\right) \tag{2}$$

where σ denotes the standard deviation of all observed pairwise distances. The adjacency matrix $A \in \mathbb{R}^{S \times S}$ is defined via threshold $\kappa = 0.1$:

$$A_{ij} = \begin{cases} W_{ij}, & \text{if } W_{ij} \geq \kappa \text{ and } i \neq j \\ 1.0, & \text{if } i = j \\ 0.0, & \text{otherwise} \end{cases} \tag{3}$$

To ensure structural connectivity, the adjacency is symmetrized ($A = \max(A, A^T)$). The resulting METR-LA graph consists of 207 nodes and 1,313 undirected edges with a mean nodal degree of 12.7.

D. Chronological Partitioning Protocol

Random k-fold cross-validation or random train-test splitting introduces catastrophic future-data leakage due to temporal autocorrelation. We enforce strict chronological partitioning:

- **Training Partition**: Earliest 70% ($t \in [1, 23990]$, 23,973 forecast origins).
- **Validation Partition**: Next 10% ($t \in [23991, 27417]$, 3,415 forecast origins).

- **Testing Partition**: Final 20% ($t \in [27418, 34272]$, 6,843 forecast origins).

Input scaling parameters (mean μ_s and standard deviation σ_s) are computed strictly across training timestamps $t \in [1, 23990]$:

$$\tilde{v}_{s,t} = \frac{v_{s,t} - \mu_s}{\sigma_s} \tag{4}$$

IV. Methodology and Formulations

A. Feature Engineering Formulations

For AML models, each observation at forecast origin t for sensor s is mapped into a structured feature vector $\mathbf{x}_{s,t}$.

1) Causal Speed Lags: To capture immediate velocity, momentum, and acceleration:

$$\mathbf{x}_{\text{lag}}(s,t) = [v_{s,t},\ v_{s,t-1},\ v_{s,t-2},\ v_{s,t-5}]^T \tag{5}$$

corresponding to lags of 0, 5, 10, and 25 minutes prior to origin t.

2) Cyclical Calendar Transformations: To capture periodic traffic oscillations across diurnal and weekly cycles without introducing artificial boundary discontinuities:

$$\phi^{\sin}_{\text{tod}}(t) = \sin\left(2\pi \cdot \tau_{\text{min}}(t)/1440\right) \tag{6}$$

$$\phi^{\cos}_{\text{tod}}(t) = \cos\left(2\pi \cdot \tau_{\text{min}}(t)/1440\right) \tag{7}$$

$$\phi^{\sin}_{\text{dow}}(t) = \sin\left(2\pi \cdot \tau_{\text{day}}(t)/7\right) \tag{8}$$

$$\phi^{\cos}_{\text{dow}}(t) = \cos\left(2\pi \cdot \tau_{\text{day}}(t)/7\right) \tag{9}$$

where $\tau_{\text{min}}(t) \in [0, 1439]$ denotes the minute of the day, and $\tau_{\text{day}}(t) \in [0, 6]$ represents the day of the week.

3) 1-Hop Spatial Neighbor Aggregation: To inject network topology into tabular regressors, we aggregate the concurrent and lagged speeds of adjacent road segments:

$$\bar{v}_{\mathcal{N}(s)}(t) = \frac{\sum_{j \in \mathcal{N}(s)} A_{sj} v_{j,t}}{\sum_{j \in \mathcal{N}(s)} A_{sj}} \tag{10}$$

where $\mathcal{N}(s) = \{j : j \neq s, A_{sj} > 0\}$.

4) Causal Rolling Moments: To quantify localized acceleration and volatility over past windows $w \in \{3, 6\}$ (15 and 30 minutes):

$$\mu^{(w)}_{s,t} = \frac{1}{w}\sum_{k=0}^{w-1} v_{s,t-k}, \quad \sigma^{(w)}_{s,t} = \sqrt{\frac{1}{w}\sum_{k=0}^{w-1}\left(v_{s,t-k} - \mu^{(w)}_{s,t}\right)^2} \tag{11}$$

B. Mathematical Formulation of Evaluation Metrics

Let $y_{s,t+h}$ denote the ground-truth observed speed for sensor s at forecast lead time $h \in \{1, 3, 6, 12\}$ steps (5, 15, 30, 60 minutes), and let $\hat{y}_{s,t+h}$ denote the corresponding model prediction. The evaluation set of valid test observations is defined as:

$$\Omega_h = \{(s,t) : t \in \mathcal{T}_{\text{test}},\ m_{s,t+h} = 1\} \tag{12}$$

We compute three standard regression metrics strictly over Ω_h:

1) Masked Mean Absolute Error (MAE):

$$\text{MAE}_h = \frac{1}{|\Omega_h|}\sum_{(s,t)\in\Omega_h} |y_{s,t+h} - \hat{y}_{s,t+h}| \tag{13}$$

2) Masked Root Mean Squared Error (RMSE):

$$\text{RMSE}_h = \sqrt{\frac{1}{|\Omega_h|}\sum_{(s,t)\in\Omega_h} (y_{s,t+h} - \hat{y}_{s,t+h})^2} \tag{14}$$

3) Masked Mean Absolute Percentage Error (MAPE):

$$\text{MAPE}_h = \frac{100\%}{|\Omega_h|}\sum_{(s,t)\in\Omega_h} \left|\frac{y_{s,t+h} - \hat{y}_{s,t+h}}{y_{s,t+h}}\right| \tag{15}$$

C. Model Architectures

1) Segment-wise Multi-Output Random Forest: For each individual sensor segment $s \in \{1, \ldots, 207\}$, an independent ensemble of randomized regression trees is trained:

$$\hat{\mathbf{y}}_{s,t} = f_s(\mathbf{x}_{s,t}) \in \mathbb{R}^4 \tag{16}$$

predicting $[y_{s,t+1}, y_{s,t+3}, y_{s,t+6}, y_{s,t+12}]^T$ concurrently. Hyperparameters are configured with 10 trees, maximum tree depth of 10, and minimum leaf sample size of 8, using up to 2,000 complete historical timestamps per sensor.

2) Pooled Sensor-Aware XGBoost: A single gradient-boosted decision tree model is pooled across all sensors for each target horizon. The objective function at boosting step m minimizes the regularized Taylor expansion:

$$\mathcal{L}^{(m)} = \sum_{i=1}^{N}\left[g_i f_m(\mathbf{x}_i) + \frac{1}{2}h_i f_m^2(\mathbf{x}_i)\right] + \gamma T + \frac{1}{2}\lambda\sum_{j=1}^{T} w_j^2 \tag{17}$$

where g_i and h_i denote first and second-order gradients. Categorical sensor indices are encoded to preserve segment heterogeneity. The model is trained with 100 boosting trees, maximum depth of 6, and learning rate $\eta = 0.08$.

3) Spatial Graph Attention Network (GAT): The deep learning baseline processes node feature tensors $\mathbf{H} \in \mathbb{R}^{B\times S\times D_{in}}$ consisting of six normalized speed lags and four cyclical temporal features ($D_{in} = 10$). Node representations are linearly projected to hidden dimension $D_h = 32$:

$$\mathbf{z}_i = \mathbf{W}_p\mathbf{h}_i \in \mathbb{R}^{D_h} \tag{18}$$

The attention coefficient α_{ij} between sensor i and neighbor $j \in \mathcal{N}(i)$ is computed via masked self-attention with edge bias b_{ij}:

$$\alpha_{ij} = \frac{\exp\left(\text{LeakyReLU}\left(\mathbf{a}_s^T\mathbf{z}_i + \mathbf{a}_t^T\mathbf{z}_j\right) + b_{ij}\right)}{\sum_{k\in\mathcal{N}(i)}\exp\left(\text{LeakyReLU}\left(\mathbf{a}_s^T\mathbf{z}_i + \mathbf{a}_t^T\mathbf{z}_k\right) + b_{ik}\right)} \tag{19}$$

where $b_{ij} = \ln(A_{ij})$ for $A_{ij} > 0$ and $-\infty$ otherwise. Messages are aggregated across neighbors and passed through LayerNorm and an ELU activation:

$$\mathbf{h}'_i = \text{ELU}\left(\text{LayerNorm}\left(\mathbf{z}_i + \sum_{j\in\mathcal{N}(i)}\alpha_{ij}\mathbf{z}_j\right)\right) \tag{20}$$

A multi-layer perceptron head maps $\mathbf{h}'_i$ to four output speed predictions. The model was trained with masked MSE loss and early stopping (best validation MSE: 0.4220 at epoch 3).

V. Experimental Results and Analysis

A. Overall Benchmark Comparison

Table I summarizes the held-out test performance across all 1,244,846 valid targets for horizons $h \in \{5, 15, 30, 60\}$ minutes.

TABLE I
Overall Held-Out Test Evaluation on METR-LA (All Periods)

Horizon	Model	MAE	RMSE	MAPE (%)	Valid Pts
5 min	Persistence	2.814	4.627	6.489	1,244,846
5 min	Random Forest	**2.519**	**4.281**	**6.327**	1,244,846
5 min	XGBoost	2.554	4.326	6.435	1,244,846
5 min	GAT	2.879	4.511	7.206	1,244,846
15 min	Persistence	3.499	6.418	8.612	1,244,834
15 min	Random Forest	**3.119**	**5.702**	**8.431**	1,244,834
15 min	XGBoost	3.239	5.873	8.846	1,244,834
15 min	GAT	3.466	5.830	9.168	1,244,834
30 min	Persistence	4.207	8.052	10.858	1,244,816
30 min	Random Forest	**3.635**	**6.706**	**10.401**	1,244,816
30 min	XGBoost	3.854	6.988	11.106	1,244,816
30 min	GAT	4.087	6.965	11.434	1,244,816
60 min	Persistence	5.381	10.345	14.535	1,244,780
60 min	Random Forest	**4.192**	**7.657**	**12.623**	1,244,780
60 min	XGBoost	4.542	7.982	13.696	1,244,780
60 min	GAT	4.898	8.227	14.328	1,244,780

B. Peak vs. Off-Peak Period Analysis

Traffic prediction during peak congestion periods is substantially more difficult due to abrupt shockwave propagation. Table II reports the partitioned evaluation for Peak (weekdays 07:00–09:59 and 16:00–18:59, 250,523 valid points) and Off-Peak (all other timestamps, 994,323 valid points).

TABLE II
Held-Out Performance Stratified by Peak and Off-Peak Periods

Period	Horizon	Model	MAE	RMSE	MAPE (%)
Peak	5 min	Persistence	3.287	5.479	10.044
Peak	5 min	Random Forest	**3.189**	**5.381**	10.712
Peak	5 min	XGBoost	3.243	5.452	10.832
Peak	5 min	GAT	3.601	5.618	11.918
Peak	15 min	Persistence	4.565	8.223	14.384
Peak	15 min	Random Forest	**4.341**	**7.580**	**15.101**
Peak	15 min	XGBoost	4.535	7.868	15.820
Peak	15 min	GAT	4.737	7.713	15.852
Peak	30 min	Persistence	5.831	10.552	19.080
Peak	30 min	Random Forest	**5.324**	**9.105**	**19.281**
Peak	30 min	XGBoost	5.808	9.612	20.926
Peak	30 min	GAT	6.001	9.487	20.818
Peak	60 min	Persistence	7.804	13.683	27.225
Peak	60 min	Random Forest	**6.374**	**10.495**	**24.259**
Peak	60 min	XGBoost	7.277	11.231	26.955
Peak	60 min	GAT	7.940	11.619	27.820
Off-peak	5 min	Persistence	2.695	4.386	5.593
Off-peak	5 min	Random Forest	**2.350**	**3.955**	**5.222**
Off-peak	5 min	XGBoost	2.381	3.992	5.328
Off-peak	5 min	GAT	2.696	4.187	6.019
Off-peak	15 min	Persistence	3.230	5.877	7.157
Off-peak	15 min	Random Forest	**2.812**	**5.121**	**6.750**
Off-peak	15 min	XGBoost	2.913	5.253	7.089
Off-peak	15 min	GAT	3.146	5.249	7.484
Off-peak	30 min	Persistence	3.797	7.288	8.787
Off-peak	30 min	Random Forest	**3.210**	**5.951**	**8.163**
Off-peak	30 min	XGBoost	3.362	6.152	8.632
Off-peak	30 min	GAT	3.605	6.169	9.070
Off-peak	60 min	Persistence	4.770	9.318	11.338
Off-peak	60 min	Random Forest	**3.643**	**6.756**	**9.691**
Off-peak	60 min	XGBoost	3.853	6.927	10.356
Off-peak	60 min	GAT	4.131	7.121	10.929

C. Key Empirical Insights

1) **Random Forest Outperforms Across All Horizons**: The segment-wise Random Forest regressor achieves the lowest MAE at all horizons (2.519 at 5 min; 4.192 at 60 min). Independent tree ensembles specialize effectively on individual segment behavior without suffering from negative interference across distant freeway corridors.
2) **XGBoost vs. Persistence**: Pooled XGBoost consistently outperforms persistence across all horizons, achieving 4.542 MAE at 60 minutes compared to 5.381 for persistence.
3) **GAT Performance Analysis**: The standalone spatial GAT improves upon naive persistence at 15, 30, and 60 minutes (4.898 vs. 5.381 MAE at 60 min). However, it does not beat Random Forest or XGBoost. Diagnostic investigation reveals that spatial attention over static flattened lag features without recurrent or temporal convolutional modeling fails to capture fine-grained sequential momentum.

VI. Limitations and Signal Simulation Readiness

A critical engineering consideration in ITS is establishing simulation readiness. Although Project 4 proposes signal control simulation, our investigation confirms significant physical data constraints:

- **Speed vs. Queue and Flow**: Loop detectors supply space-mean speed (v). Signal timing optimization (e.g. via Webster's formula or reinforcement learning [9]) requires vehicular arrival rates (flow q) and queue lengths (k). Deriving queue lengths from speed alone without signal phase green times and saturation flow constants is physically ill-posed.
- **Symmetric Adjacency**: The Gaussian road distance matrix is undirected ($A_{ij} = A_{ji}$), whereas vehicular propagation is strictly directional (upstream to downstream).

VII. Conclusion and Phase 2 Outlook

This study established a rigorous, leakage-free benchmark for short-term traffic speed forecasting on the METR-LA network. Applied machine learning models, specifically segment-wise Random Forests, demonstrated superior predictive accuracy across all horizons and operating periods, proving that localized non-parametric ensembles remain exceptionally strong baselines.

In **Phase 2**, we will leverage these empirical insights to develop a **Residual Spatial Hybrid Architecture**. By using the Random Forest/XGBoost ensemble to predict base localized trends and training a recurrent Graph Neural Network on the residual spatial errors ($e_{s,t} = y_{s,t} - \hat{y}_{s,t}^{\text{AML}}$), we aim to unlock

network-wide spatial message passing and capture propagating congestion shockwaves.

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